CMSC 2621
Lab Exercise 4

Due: See the assignment folder for the due date.

Overview:

In this exercise, you will simulate a zombie apocalypse scenario where you must manage your resources and actions using a stack. You will implement the stack data structure in C++ and use it to handle various survival tasks. By the end of this exercise, you will have a solid understanding of how stacks work and how to implement them in C++.

Requirements:

1. Download the ZombieApocalypse.cpp file that's linked in the D2L assignment. Follow the instructions in the comments and update only the locations in the code that have a "CHANGE_HERE"
2. Submit the source code and a document of a screenshot in a zip file to the D2L assignment. (Please see the instructions in the end of this document)

Submission Requirements:

You'll be submitting a zip file that contains the files below:

1. The source code file
2. A Word document containing screenshots of your source code and the full results of executing your program.

After creating a zip file containing all the required files, submit the zip file to the assignment folder on D2L.

Screenshot Requirements:

1. The screenshot(s) need to be clear enough to read the source code.
2. If the running results are too long to be captured, use multiple screenshots.
3. The running result(s) need to be captured in the screenshot(s) completely. If the running results are not complete, i.e. just partial results are captured in the screenshot(s), you may also receive a points deduction, depending on how complete the results in the screenshots are.

Assignment Grading:

1. If the program fails to be compiled using a compiler that supports C++ 20, a maximum grade of 40% will be given.
2. If any requirements are not met, for each unfulfillment, a corresponding penalty will be given.
3. Completeness of the lab assignment will also be considered.

Example Output:

```
Added Shotgun to backpack.
Added Medkit to backpack.
Added Ammo to backpack.
Added Water to backpack.
Added Food to backpack.
Stack is full! Can't add more items.
Using Food to survive.
Using Water to survive.
Using Ammo to survive.
Peeking at the top item: Medkit
Using Medkit to survive.
Using Shotgun to survive.
Stack is empty! Can't pop items.
Added Shotgun to magic backpack.
Added Medkit to magic backpack.
Added Ammo to magic backpack.
Added Water to magic backpack.
Added Food to magic backpack.
Added Grenade to magic backpack.
Added Axe to magic backpack.
Added Bow to magic backpack.
Added Bat to magic backpack.
Added Medicine to magic backpack.
Added Rifle to magic backpack.
Added Shield to magic backpack.
Using Shield to survive.
Using Rifle to survive.
Using Medicine to survive.
Added Item Item0 to magic backpack.
Magic backpack capacity is 20
```

```
Magic backpack capacity is 20
Added Item Item1 to magic backpack.
Magic backpack capacity is 20
Added Item Item2 to magic backpack.
Magic backpack capacity is 20
Added Item Item3 to magic backpack.
Magic backpack capacity is 20
Added Item Item4 to magic backpack.
Magic backpack capacity is 20
Added Item Item5 to magic backpack.
Magic backpack capacity is 20
Added Item Item6 to magic backpack.
Magic backpack capacity is 20
Added Item Item7 to magic backpack.
Magic backpack capacity is 20
Added Item Item8 to magic backpack.
Magic backpack capacity is 20
Added Item Item9 to magic backpack.
Magic backpack capacity is 20
Added Item Item10 to magic backpack.
Magic backpack capacity is 20
Added Item Item11 to magic backpack.
Magic backpack capacity is 40
Added Item Item12 to magic backpack.
Magic backpack capacity is 40
Added Item Item13 to magic backpack.
Magic backpack capacity is 40
```

```
Added Item Item14 to magic backpack.
Magic backpack capacity is 40
Added Item Item15 to magic backpack.
Magic backpack capacity is 40
Added Item Item16 to magic backpack.
Magic backpack capacity is 40
Added Item Item17 to magic backpack.
Magic backpack capacity is 40
Added Item Item18 to magic backpack.
Magic backpack capacity is 40
Added Item Item19 to magic backpack.
Magic backpack capacity is 40
Added Item Item20 to magic backpack.
Magic backpack capacity is 40
Added Item Item21 to magic backpack.
Magic backpack capacity is 40
Added Item Item22 to magic backpack.
Magic backpack capacity is 40
Added Item Item23 to magic backpack.
Magic backpack capacity is 40
Added Item Item24 to magic backpack.
Magic backpack capacity is 40
Added Item Item25 to magic backpack.
Magic backpack capacity is 40
Added Item Item26 to magic backpack.
Magic backpack capacity is 40
Added Item Item27 to magic backpack.
Magic backpack capacity is 40
```

```
Added Item Item28 to magic backpack.
Magic backpack capacity is 40
Added Item Item29 to magic backpack.
Magic backpack capacity is 40
Added Item Item30 to magic backpack.
Magic backpack capacity is 40
Added Item Item31 to magic backpack.
Magic backpack capacity is 80
Added Item Item32 to magic backpack.
Magic backpack capacity is 80
Added Item Item33 to magic backpack.
Magic backpack capacity is 80
Added Item Item34 to magic backpack.
Magic backpack capacity is 80
Added Item Item35 to magic backpack.
Magic backpack capacity is 80
Added Item Item36 to magic backpack.
Magic backpack capacity is 80
Added Item Item37 to magic backpack.
Magic backpack capacity is 80
Added Item Item38 to magic backpack.
Magic backpack capacity is 80
Added Item Item39 to magic backpack.
Magic backpack capacity is 80
```